

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		[Atoms of] the same <u>element</u> with the same number of protons (1) different number of neutrons or different mass number or different nucleon number (1) Don't accept same number of protons and neutrons or different number of protons and neutrons	2			2		
		(ii)		Due to an imbalance between number of protons and neutrons or they have unstable nuclei	1			1		
	(b)			Only alpha blocked by the smoke [so caesium not suitable] (1) ORA It has a much longer half-life [than radium] (1) ORA so it will not need frequent replacing or lasts long enough (1)		3		3		
	(c)	(i)		Initial activity = $75 \times 16 = 1200$ [MBq] (1) Sequence of halving from 1200 shown i.e. 1200 - 600 - 300 - 150 - 75 - 37.5 leading to 5 half-lives (1) $5 \times 60 = 300$ [days] (1) Answer only of $\frac{1}{2}$ award 1 mark		3		3	3	
		(ii)		Gamma is [very] penetrating [so could expose people] (1) and can damage <u>cells</u> or is [weakly] ionising (1)		2		2		
				Question 5 total	3	8	0	11	3	0